In [ ]:
```
EXPERIMENT 6 – DATA VISUALIZATION WITH SEABORN
PIN NO.: 241UDS0101

 AIM :-
To visualize and analyze a dataset using Seaborn plots such as scatter
plots, KDE plots, histograms, count plots, bar plots, boxplots, strip plots,
joint plots, and heatmaps.
```

In [ ]:
```
 Prerequisites & Requirements
1. Python installed
2. A CSV dataset
3. Jupyter Notebook or VS Code
4. Basic knowledge of Pandas, Matplotlib, NumPy, and Seaborn
```

In [ ]:
```
## Step 1 – Import the Required Libraries & Dataset
```

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

In [2]:
```
# Load the dataset
df_241UDS0101 = pd.read_csv("Experiment_6_241UDS0101_Renewable_Energy.csv")

# Display the dataset
df_241UDS0101.head()
```

Out[2]:

| | city | date | energy_mwh | state | energy_source | PIN |
|---|---|---|---|---|---|---|
| **0** | Vijayawada | 2022-01-15 | 18000 | Andhra Pradesh | Solar | 241UDS0101 |
| **1** | Kurnool | 2022-04-15 | 20500 | Andhra Pradesh | Wind | 241UDS0101 |
| **2** | Kadapa | 2022-07-15 | 22200 | Andhra Pradesh | Hydro | 241UDS0101 |
| **3** | Nellore | 2022-10-15 | 19800 | Andhra Pradesh | Solar | 241UDS0101 |
| **4** | Vijayawada | 2023-01-15 | 22660 | Andhra Pradesh | Solar | 241UDS0101 |

In [ ]:
```
## Step 2 – Info of the Dataset
```

In [3]:
```
df_241UDS0101.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 80 entries, 0 to 79
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype 
---  ------         --------------  ----- 
 0   city           80 non-null     object
 1   date           80 non-null     object
 2   energy_mwh     80 non-null     int64 
 3   state          80 non-null     object
 4   energy_source  80 non-null     object
 5   PIN            80 non-null     object
dtypes: int64(1), object(5)
memory usage: 3.9+ KB
```

In [ ]:
```
## Step 3 — State-wise Data Points
```

In [4]:
```
df_241UDS0101.value_counts("state")
```

Out[4]:
```
state
Andhra Pradesh    16
Gujarat           16
Karnataka         16
Rajasthan         16
Tamil Nadu        16
Name: count, dtype: int64
```

In [ ]:
```
## Step 4 — Convert Date Column to Datetime
```

In [5]:
```
df_241UDS0101["date"] = pd.to_datetime(df_241UDS0101["date"])

# Create a year column for heatmap analysis
df_241UDS0101["year_241UDS0101"] = df_241UDS0101["date"].dt.year

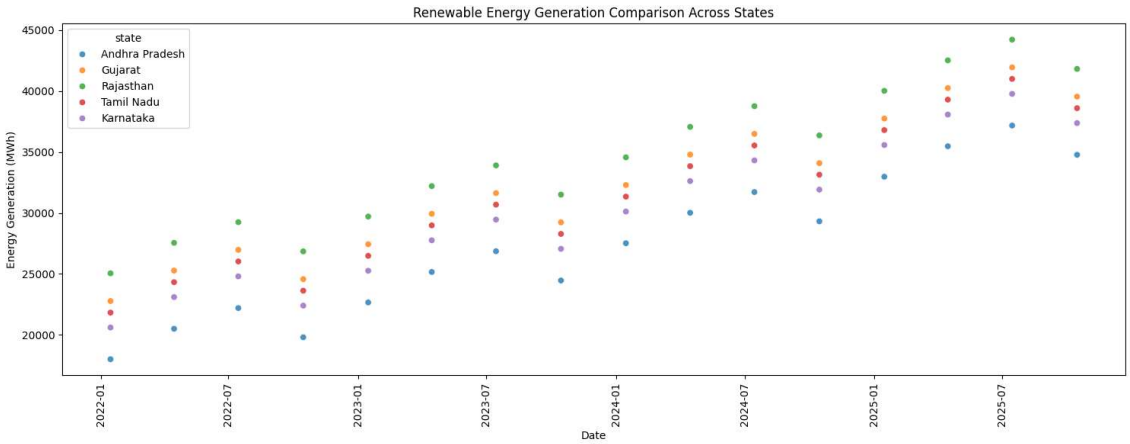
df_241UDS0101.head()
```

Out[5]:

| | city | date | energy_mwh | state | energy_source | PIN | year_241UDS0101 |
|---|---|---|---|---|---|---|---|
| **0** | Vijayawada | 2022-01-15 | 18000 | Andhra Pradesh | Solar | 241UDS0101 | 2022 |
| **1** | Kurnool | 2022-04-15 | 20500 | Andhra Pradesh | Wind | 241UDS0101 | 2022 |
| **2** | Kadapa | 2022-07-15 | 22200 | Andhra Pradesh | Hydro | 241UDS0101 | 2022 |
| **3** | Nellore | 2022-10-15 | 19800 | Andhra Pradesh | Solar | 241UDS0101 | 2022 |
| **4** | Vijayawada | 2023-01-15 | 22660 | Andhra Pradesh | Solar | 241UDS0101 | 2023 |

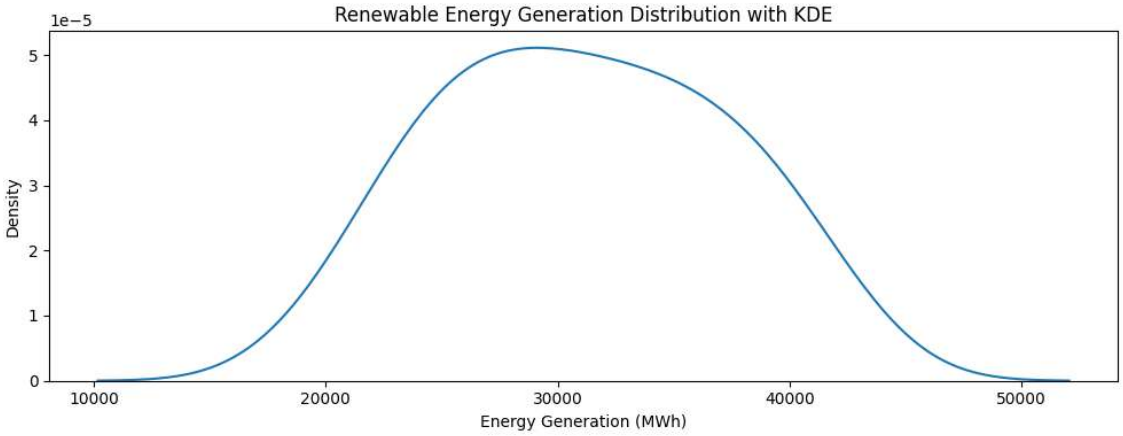
In [ ]:
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## Step 5 — Visualize Data Using Various Plots
```

In [6]:
```
# Scatter Plot: renewable energy generation across states over time
plt.figure(figsize=(15, 6))
sns.scatterplot(
    x="date",
    y="energy_mwh",
    hue="state",
    data=df_241UDS0101,
    alpha=0.8
)
plt.title("Renewable Energy Generation Comparison Across States")
plt.xticks(rotation=90)
plt.xlabel("Date")
plt.ylabel("Energy Generation (MWh)")
plt.tight_layout()
plt.show()
```



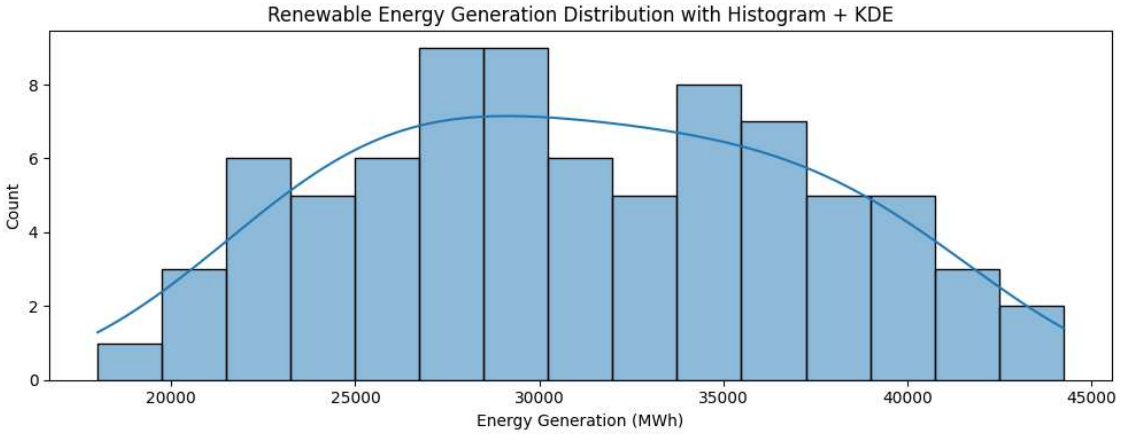

In [ ]:
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## Step 6 — Histograms & KDE
```

In [7]:
```
# KDE Plot: energy-generation distribution
plt.figure(figsize=(10, 4))
sns.kdeplot(data=df_241UDS0101, x="energy_mwh")
plt.title("Renewable Energy Generation Distribution with KDE")
plt.xlabel("Energy Generation (MWh)")
plt.ylabel("Density")
plt.tight_layout()
plt.show()
```



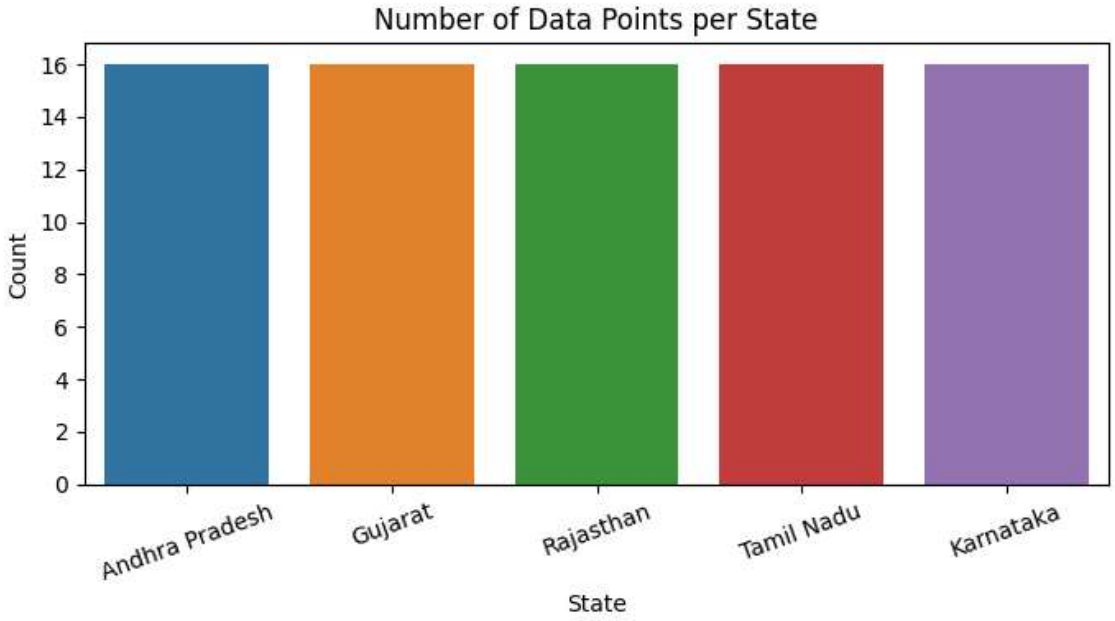

In [ ]:
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## Step 7 — Histogram + KDE
```

In [8]:
```
plt.figure(figsize=(10, 4))
sns.histplot(
    data=df_241UDS0101,
    x="energy_mwh",
    bins=15,
    kde=True
)
plt.title("Renewable Energy Generation Distribution with Histogram + KDE")
plt.xlabel("Energy Generation (MWh)")
plt.ylabel("Count")
plt.tight_layout()
plt.show()
```



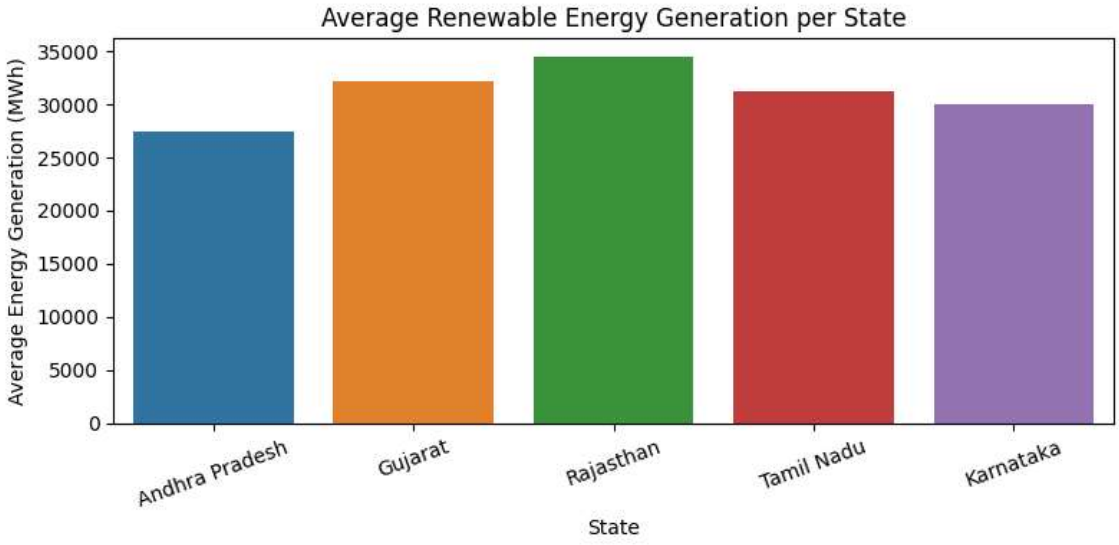
In [ ]:
```
## Step 8 – Count Plot
```

In [9]:
```
plt.figure(figsize=(7, 4))
sns.countplot(
    x="state",
    data=df_241UDS0101,
    hue="state",
    legend=False
)
plt.title("Number of Data Points per State")
plt.xlabel("State")
plt.ylabel("Count")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```



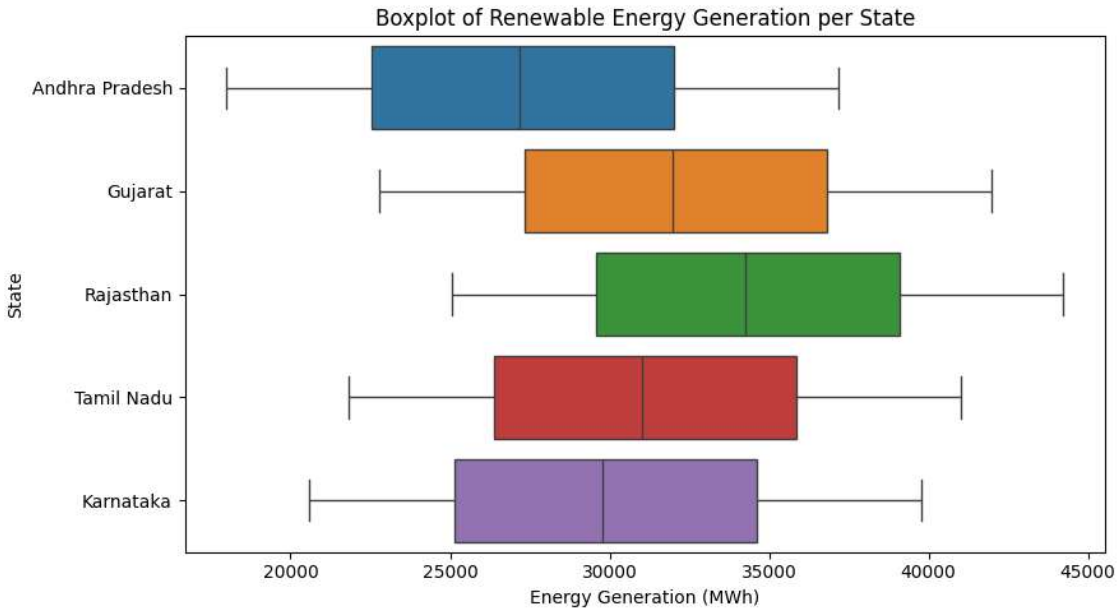

In [ ]:
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## Step 9 – Bar Plot
```

In [10]:
```
plt.figure(figsize=(8, 4))
sns.barplot(
    x="state",
    y="energy_mwh",
    data=df_241UDS0101,
    hue="state",
    estimator=np.mean,
    errorbar=None,
    legend=False
)
plt.title("Average Renewable Energy Generation per State")
plt.xlabel("State")
plt.ylabel("Average Energy Generation (MWh)")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```



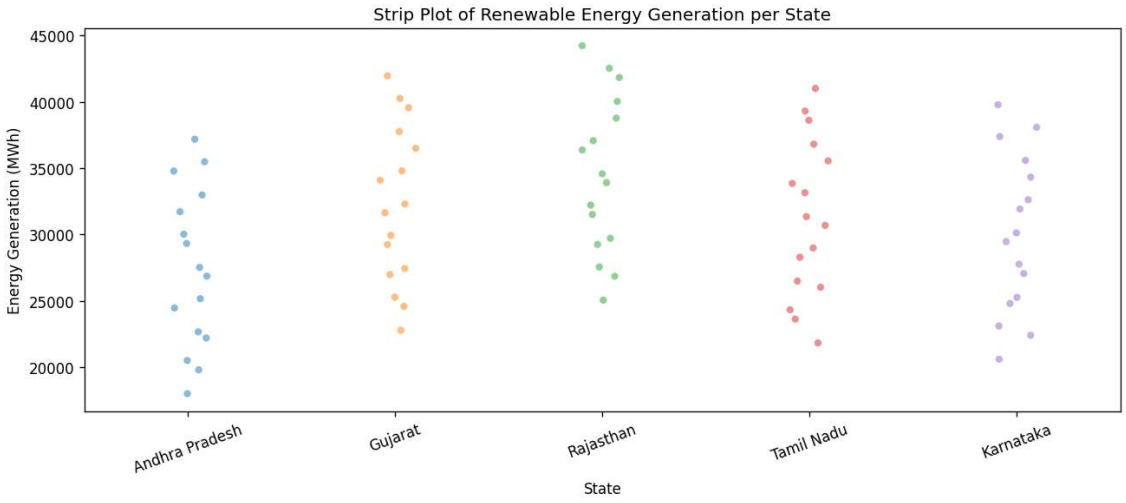

In [ ]:
```
## Step 10 — Boxplot
```

In [11]:
```
plt.figure(figsize=(9, 5), dpi=100)
sns.boxplot(
    x="energy_mwh",
    y="state",
    data=df_241UDS0101,
    hue="state",
    legend=False
)
plt.title("Boxplot of Renewable Energy Generation per State")
plt.xlabel("Energy Generation (MWh)")
plt.ylabel("State")
plt.tight_layout()
plt.show()
```



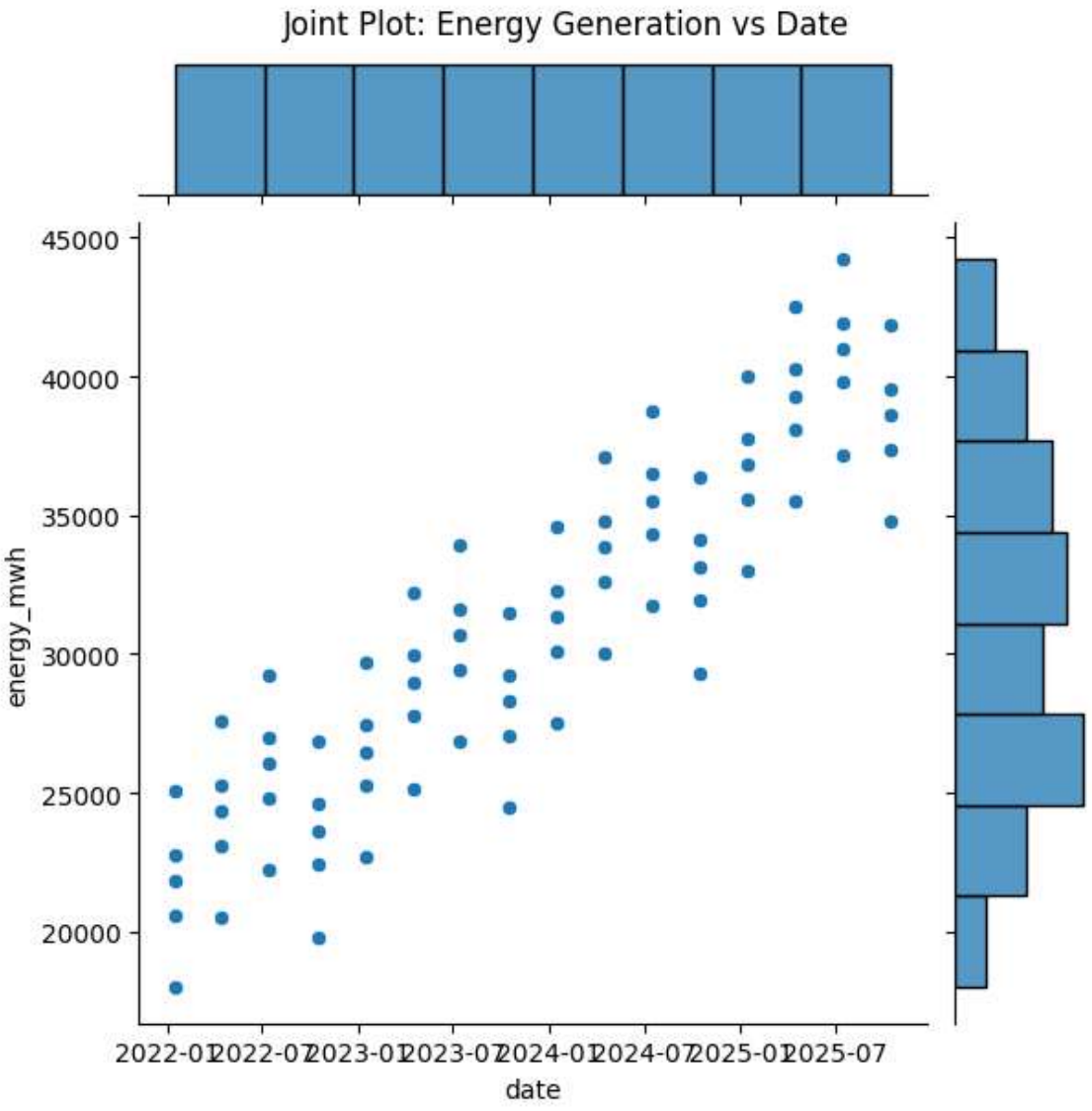

In [ ]:
```
## Step 11 — Swarm / Strip Plot
```

In [12]:
```
plt.figure(figsize=(11, 5), dpi=120)
sns.stripplot(
    x="state",
    y="energy_mwh",
    data=df_241UDS0101,
    hue="state",
    alpha=0.5,
    legend=False
)
plt.title("Strip Plot of Renewable Energy Generation per State")
plt.xlabel("State")
plt.ylabel("Energy Generation (MWh)")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```




In [ ]:
```
## Step 12 — Joint Plot
```

In [13]:
```
sns.jointplot(
    x="date",
    y="energy_mwh",
    data=df_241UDS0101,
    kind="scatter"
)
plt.suptitle("Joint Plot: Energy Generation vs Date", y=1.02)
plt.show()
```




In [ ]:
```
## Step 13 – Heat Map Preparation
```

In [14]:
```
# Average energy generation per year per state
yearly_avg_241UDS0101 = df_241UDS0101.groupby(
    ["year_241UDS0101", "state"]
)["energy_mwh"].mean()

yearly_avg_241UDS0101
```

Out[14]:
```
year_241UDS0101  state
2022             Andhra Pradesh    20125.0
                 Gujarat           24900.0
                 Karnataka         22725.0
                 Rajasthan         27175.0
                 Tamil Nadu        23950.0
2023             Andhra Pradesh    24785.0
                 Gujarat           29560.0
                 Karnataka         27385.0
                 Rajasthan         31835.0
                 Tamil Nadu        28610.0
2024             Andhra Pradesh    29645.0
                 Gujarat           34420.0
                 Karnataka         32245.0
                 Rajasthan         36695.0
                 Tamil Nadu        33470.0
2025             Andhra Pradesh    35105.0
                 Gujarat           39880.0
                 Karnataka         37705.0
                 Rajasthan         42155.0
                 Tamil Nadu        38930.0
Name: energy_mwh, dtype: float64
```

In [ ]:
```
## Step 14 — Using unstack()
```

In [15]:
```
# Reshape the result: each row = year, each column = state
yearly_avg_241UDS0101 = df_241UDS0101.groupby(
    ["year_241UDS0101", "state"]
)["energy_mwh"].mean().unstack()

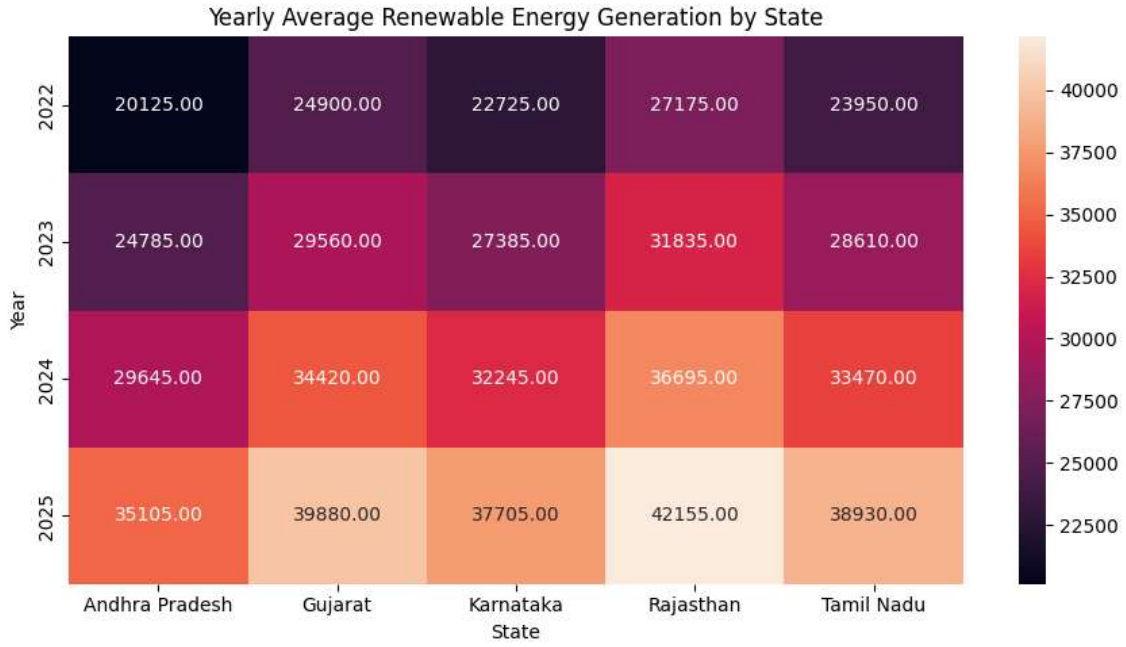
yearly_avg_241UDS0101.head()
```

Out[15]:

| state | Andhra Pradesh | Gujarat | Karnataka | Rajasthan | Tamil Nadu |
|---|---|---|---|---|---|
| year_241UDS0101 | | | | | |
| **2022** | 20125.0 | 24900.0 | 22725.0 | 27175.0 | 23950.0 |
| **2023** | 24785.0 | 29560.0 | 27385.0 | 31835.0 | 28610.0 |
| **2024** | 29645.0 | 34420.0 | 32245.0 | 36695.0 | 33470.0 |
| **2025** | 35105.0 | 39880.0 | 37705.0 | 42155.0 | 38930.0 |

In [ ]:
```
## Step 15 — Heatmap
```

In [16]:
```
plt.figure(figsize=(9, 5), dpi=100)
sns.heatmap(
    yearly_avg_241UDS0101,
    fmt=".2f",
    annot=True
)
plt.title("Yearly Average Renewable Energy Generation by State")
plt.xlabel("State")
plt.ylabel("Year")
plt.tight_layout()
plt.show()
```




In [ ]:
```
RESULT :-
Various data-visualization techniques using Seaborn were implemented to
analyze renewable energy generation across states and years.
```